Orbit & sinus relations

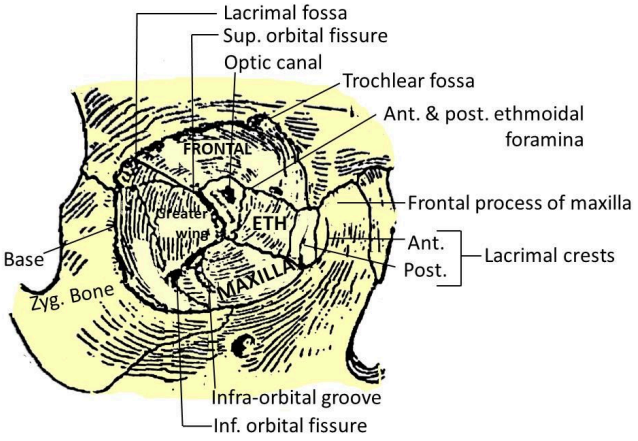
The orbit as a neighbour of every sinus: walls, apertures, vessels, the lacrimal pathway and the routes by which sinus disease reaches the eye.

1. Bony orbit
2. Orbital septum and periosteum
3. Chandler classification of orbital complications
4. Vessels and nerves at risk in sinus surgery
5. Cavernous sinus
6. Lacrimal drainage
7. On imaging
8. In theatre

01 Bony orbit

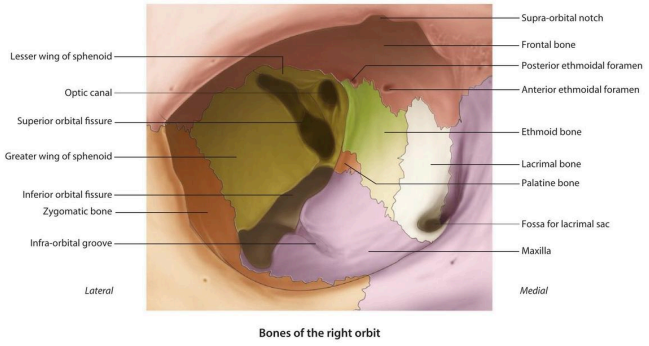
- Pyramid with the apex at the **optic canal** and base at the orbital margin. Volume about 30 mL.
- **Roof:** orbital plate of frontal bone (frontal sinus and anterior cranial fossa above) and lesser wing of sphenoid.
- **Medial wall:** frontal process of maxilla, lacrimal bone, **lamina papyracea** (ethmoid) and body of sphenoid. Thinnest wall.
- **Floor:** orbital surface of maxilla, zygomatic and palatine bones; the **infraorbital groove and canal** cross it. **Maxillary sinus** below.
- **Lateral wall:** zygomatic bone and greater wing of sphenoid. Thickest wall.
- **Openings:** optic canal (II, ophthalmic artery), superior orbital fissure, inferior orbital fissure (V2, zygomatic nerve, inferior ophthalmic vein branch to pterygoid plexus), anterior and posterior ethmoidal foramina.

CLINICAL PEARL Blow-out fractures break the floor posteromedial to the infraorbital canal or the lamina papyracea; entrapment of inferior rectus causes diplopia on upward gaze.



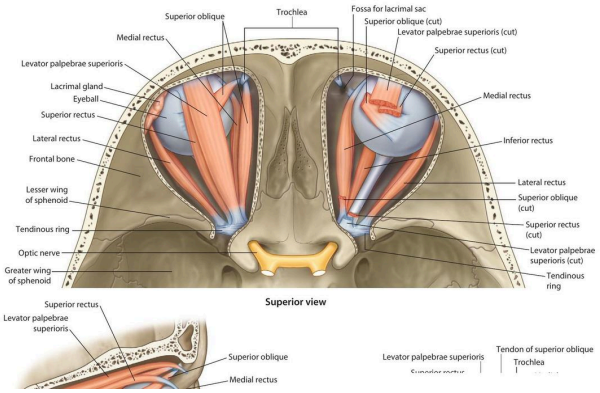
Bony orbit: walls, fissures and foramina

Dr. Sameh Doss handout, p. 90



Bones of the right orbit

Gray's Atlas of Anatomy, 2nd ed., p. 508

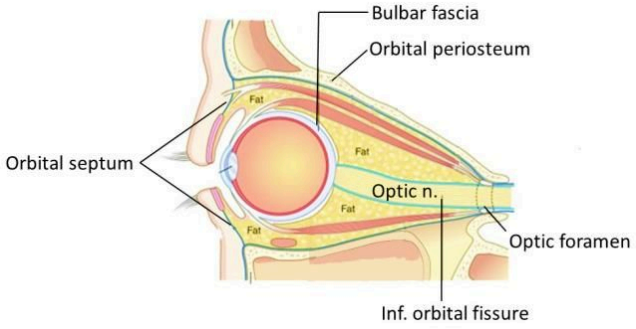


Muscles of the eyeball (superior and lateral views)

Gray's Atlas of Anatomy, 2nd ed., p. 512

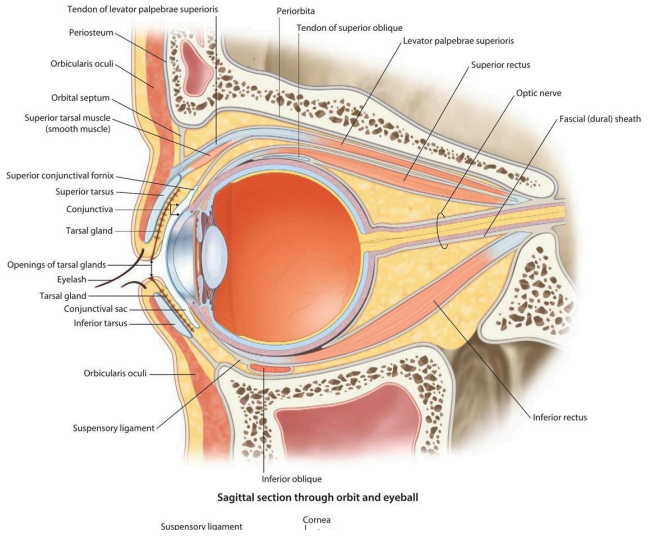
02 Orbital septum and periosteum

- **Periosteum (periorbita)** lines the orbit; loosely attached except at the sutures, fissures and margin.
- **Orbital septum** runs from the orbital margin to the tarsal plates and divides **preseptal** (eyelid) from **postseptal** (orbit proper) compartments.
- Sinus infection usually reaches the orbit across the **lamina papyracea** or through **valveless ethmoidal veins**, and collects first between the bone and periorbita (subperiosteal abscess).



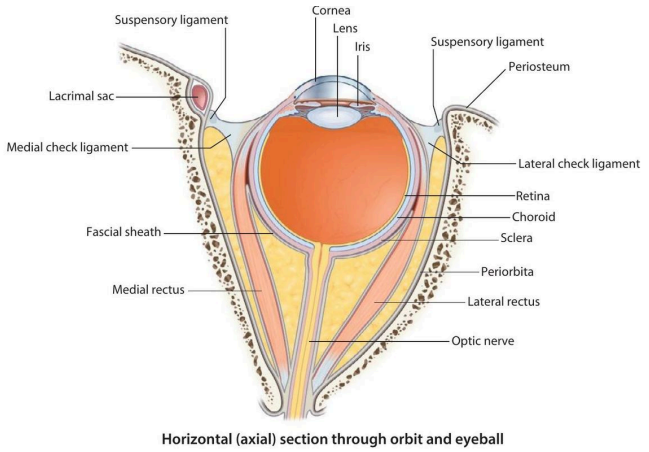
Orbital septum and periosteum

Dr. Sameh Doss handout, p. 101



Sagittal section through orbit and eyelid: orbital septum and periorbita

Gray's Atlas of Anatomy, 2nd ed., p. 509



Horizontal (axial) section through orbit and eyeball

Gray's Atlas of Anatomy, 2nd ed., p. 509

03 Chandler classification of orbital complications

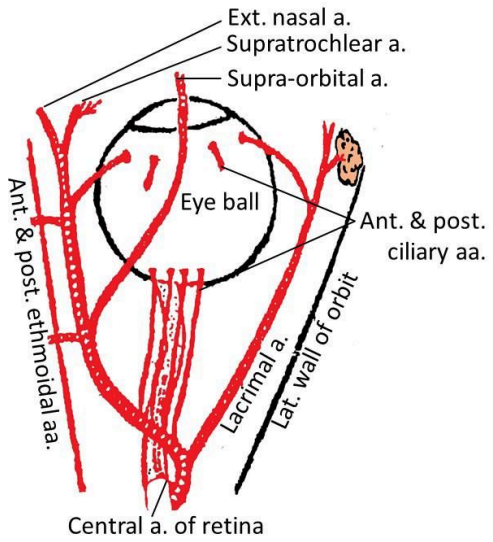
- **I. Preseptal cellulitis:** lid swelling, normal vision and eye movement.
- **II. Orbital cellulitis:** diffuse orbital oedema; proptosis, chemosis, painful or reduced movement.
- **III. Subperiosteal abscess:** pus between the lamina papyracea and periorbita; displaces the globe down and out.
- **IV. Orbital abscess:** pus within the orbital fat; severe proptosis, ophthalmoplegia, visual loss.
- **V. Cavernous sinus thrombosis:** bilateral signs, III, IV, V1, V2, VI palsies, sepsis.
- Red flags for urgent CT and drainage: reduced acuity, loss of colour vision (red desaturation), afferent pupillary defect, ophthalmoplegia.

> **CLINICAL PEARL** Most orbital complications in children come from the ethmoid sinuses.

04 Vessels and nerves at risk in sinus surgery

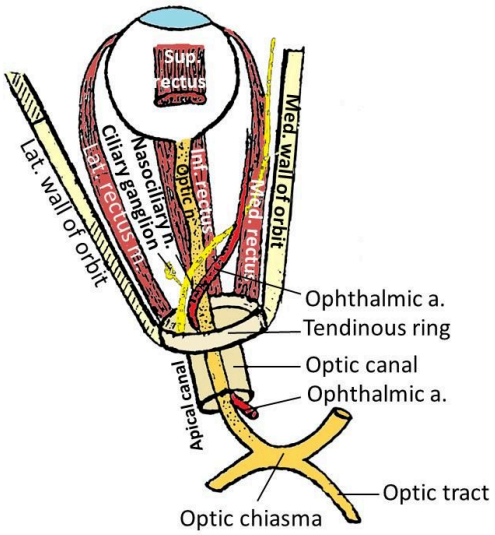
- **Anterior ethmoidal artery** leaves the orbit through the anterior ethmoidal foramen, crosses the ethmoid roof (often in a mesentery) behind the frontal recess, and enters the anterior cranial fossa.
- **Posterior ethmoidal foramen** lies about 6 mm (range 4–7 mm) in front of the optic canal.
- **Rule of 24-12-6 (approximate):** anterior lacrimal crest → anterior ethmoidal foramen 24 mm → posterior ethmoidal foramen 12 mm → optic canal 6 mm.
- **Optic nerve** can bulge into the posterior ethmoid (Onodi cell) or sphenoid sinus and may be dehiscent.
- **Ophthalmic veins** drain back to the cavernous sinus and down to the pterygoid plexus; they are valveless.

> **CLINICAL PEARL** A retrobulbar haematoma during FESS needs immediate lateral canthotomy and cantholysis; vision can be lost within about 90 minutes of ischaemia.



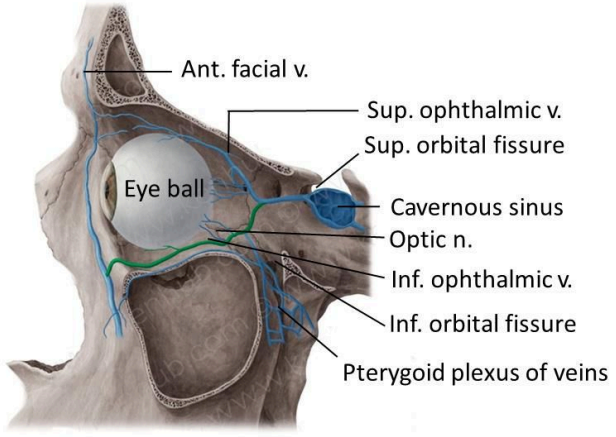
Ophthalmic artery and the anterior and posterior ethmoidal arteries

Dr. Sameh Doss handout, p. 100



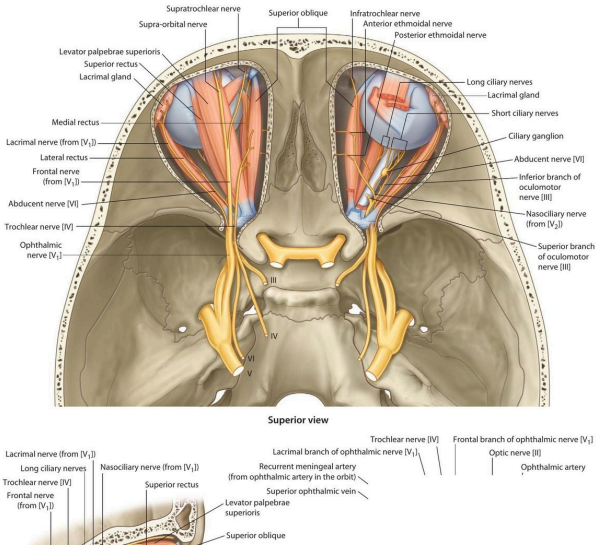
Optic canal, ophthalmic artery and the tendinous ring

Dr. Sameh Doss handout, p. 95



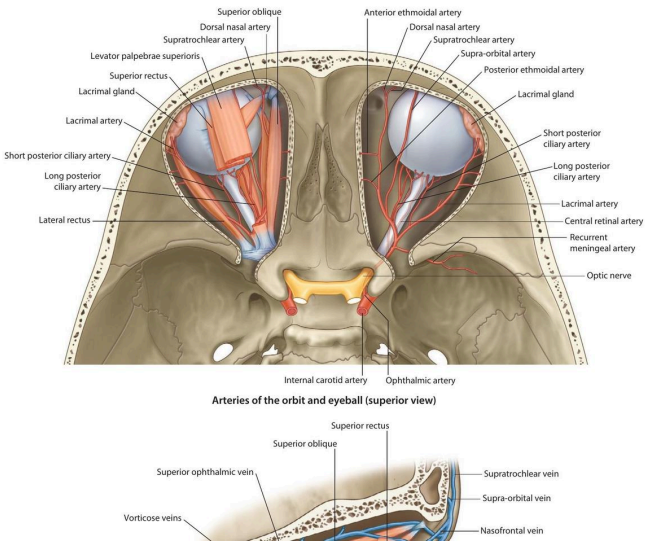
Orbital veins draining to the cavernous sinus and pterygoid plexus

Dr. Sameh Doss handout, p. 101



Innervation of the orbit (superior and lateral views)

Gray's Atlas of Anatomy, 2nd ed., p. 513



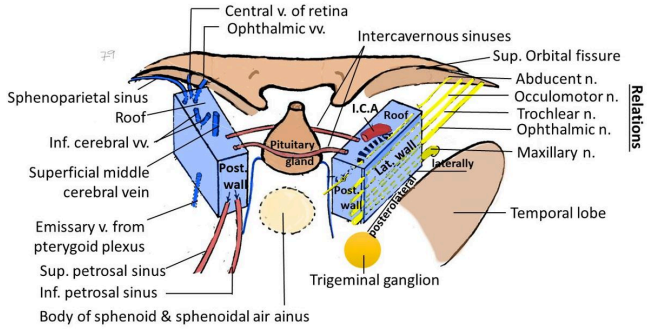
Arteries and veins of the orbit: ophthalmic and ethmoidal arteries

Gray's Atlas of Anatomy, 2nd ed., p. 515

05 Cavernous sinus

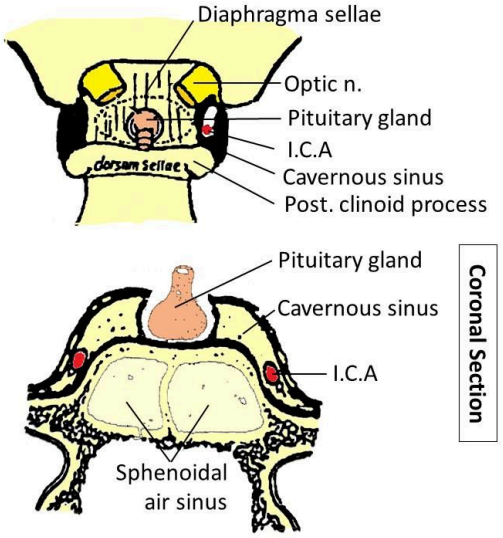
- Venous space on each side of the sphenoid body, from the superior orbital fissure to the apex of the petrous bone.
- **Lateral wall (top to bottom): III, IV, V1, V2**.
- **Inside the sinus:** the **ICA** with its sympathetic plexus and the **abducens nerve (VI)** lateral to it.
- **Tributaries:** superior and inferior ophthalmic veins, sphenoparietal sinus, superficial middle cerebral vein. **Drains** to the superior and inferior petrosal sinuses; connects to the pterygoid plexus through emissary veins.

> **CLINICAL PEARL** VI is the nerve most often affected first in cavernous sinus thrombosis because it floats free inside the sinus.



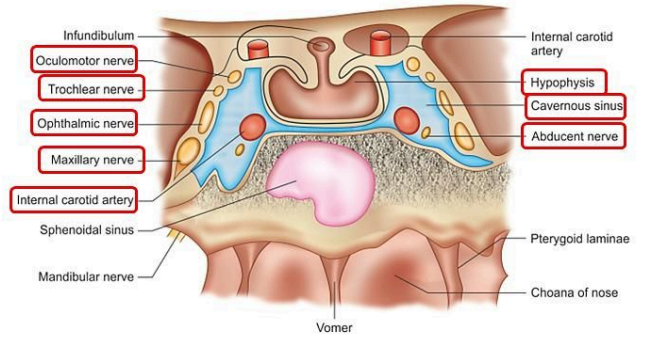
Cavernous sinus: contents and lateral wall

Dr. Sameh Doss handout, p. 85



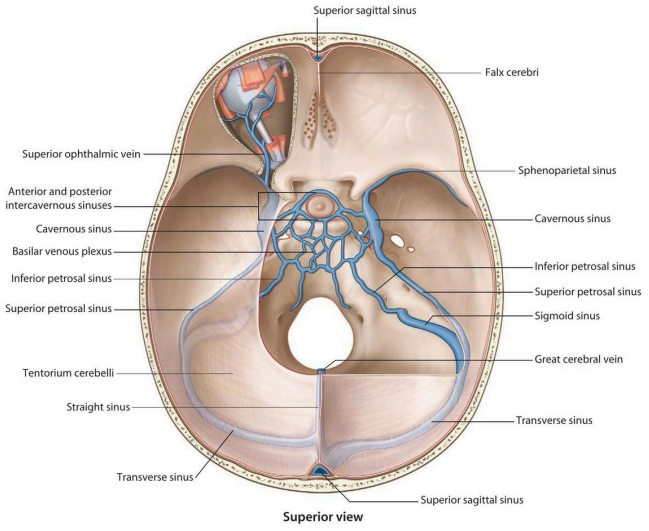
Cavernous sinus beside the sphenoid sinus, coronal view

Dr. Sameh Doss handout, p. 88



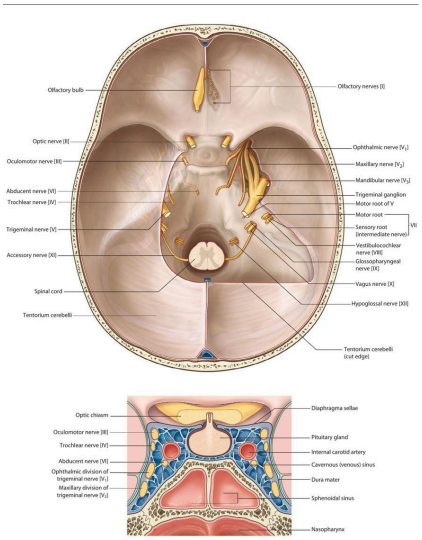
Coronal section: cavernous sinus beside the sphenoid

ENT Steps Anatomy, p. 155



Dural venous sinuses (superior view): cavernous and petrosal sinuses

Gray's Atlas of Anatomy, 2nd ed., p. 493



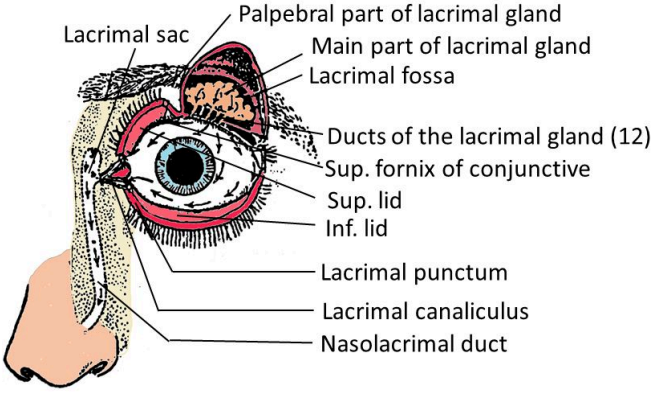
Cranial nerves in the cranial cavity, with coronal section through the cavernous sinus

Gray's Atlas of Anatomy, 2nd ed., p. 498

06 Lacrimal drainage

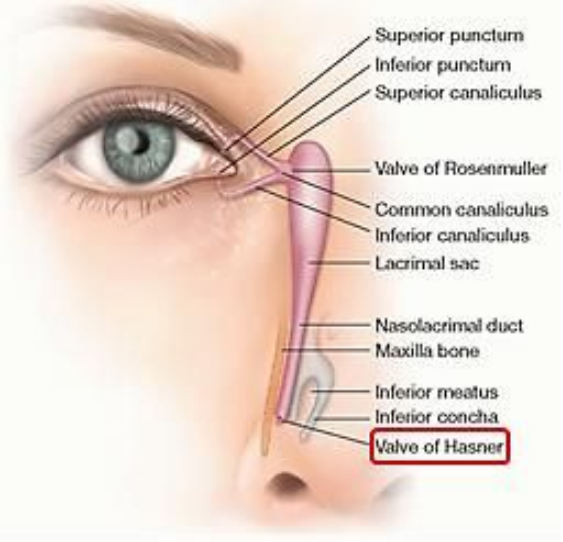
- Tears drain through the **puncta** → **canaliculi** (vertical 2 mm, horizontal 8 mm) → **common canaliculus** → **lacrimal sac** (in its fossa between the anterior and posterior lacrimal crests) → **nasolacrimal duct** (about 12–18 mm) → **inferior meatus**.
- **Valve of Hasner** guards the lower end; **valve of Rosenmüller** guards the common canaliculus entry.
- The sac lies just in front of the uncinate process attachment and the anterior end of the middle turbinate (the axilla).

CLINICAL PEARL Endoscopic DCR opens the sac into the nose just in front of the middle turbinate axilla, removing the frontal process of maxilla and lacrimal bone.



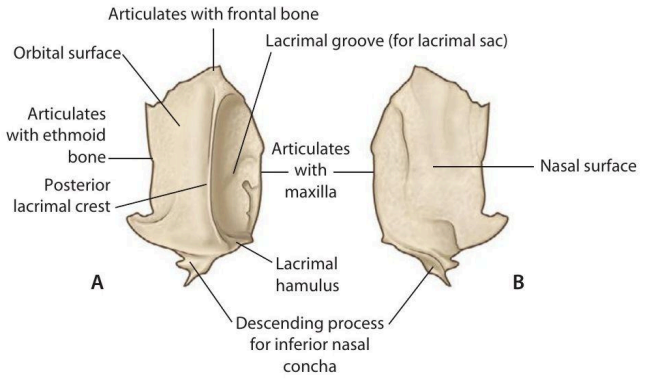
Lacrimal apparatus

Dr. Sameh Doss handout, p. 92



Lacrimal drainage: canaliculi, sac, nasolacrimal duct and Hasner's valve

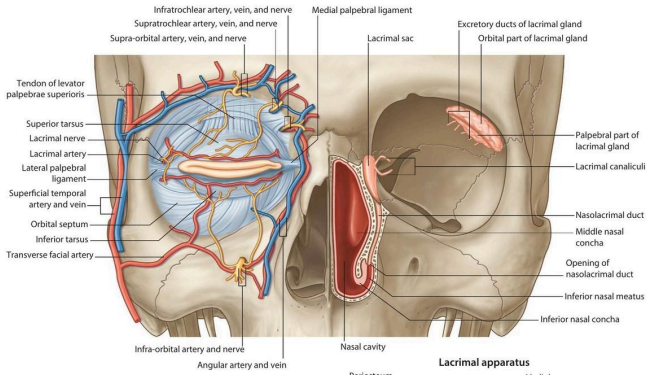
ENT Steps Anatomy, p. 100



Right lacrimal bone
A. Lateral view B. Medial view

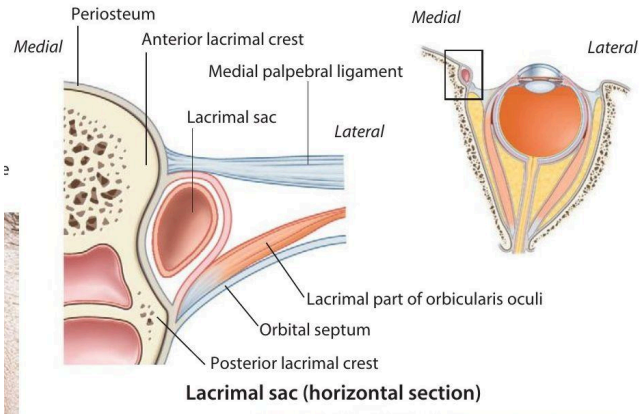
Right lacrimal bone: lateral and medial views

Gray's Atlas of Anatomy, 2nd ed., p. 486



Lacrimal apparatus and vessels and nerves of the eyelids

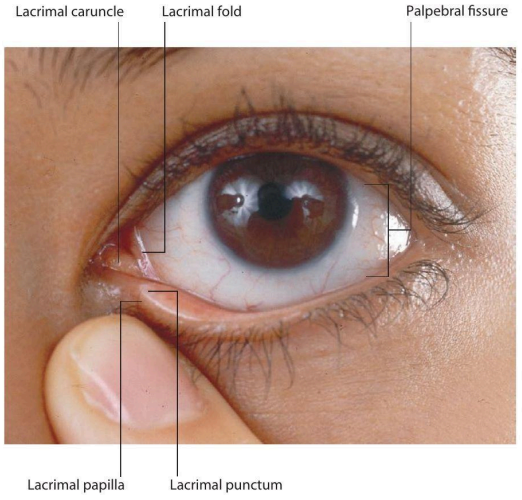
Gray's Atlas of Anatomy, 2nd ed., p. 510



Lacrimal sac (horizontal section)

Lacrimal sac (horizontal section)

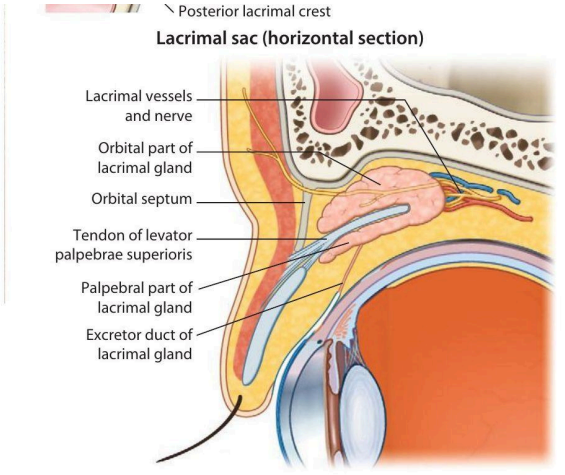
Gray's Atlas of Anatomy, 2nd ed., p. 510



Lacrimal papilla and punctum of left eye

Lacrimal papilla and punctum

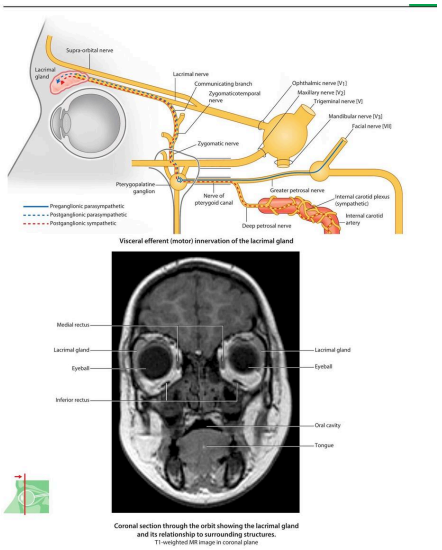
Gray's Atlas of Anatomy, 2nd ed., p. 510



Lacrimal gland and levator palpebrae superioris (parasagittal section)

Lacrimal gland and levator palpebrae superioris (parasagittal section)

Gray's Atlas of Anatomy, 2nd ed., p. 510

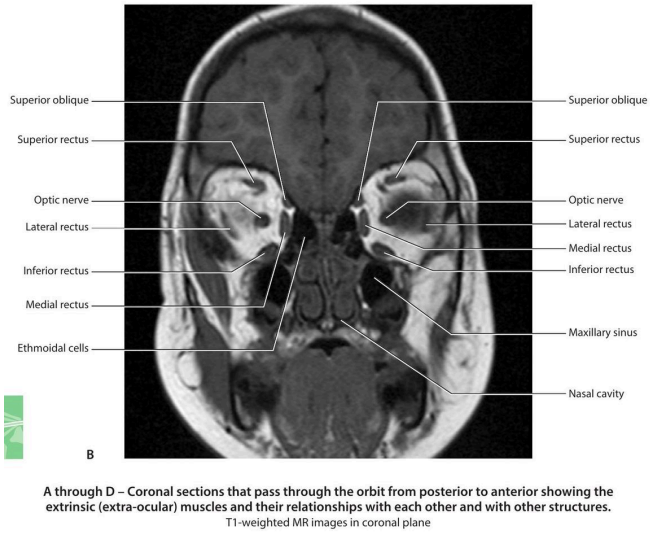


Innervation of the lacrimal gland, with coronal MRI through the orbit

Gray's Atlas of Anatomy, 2nd ed., p. 511

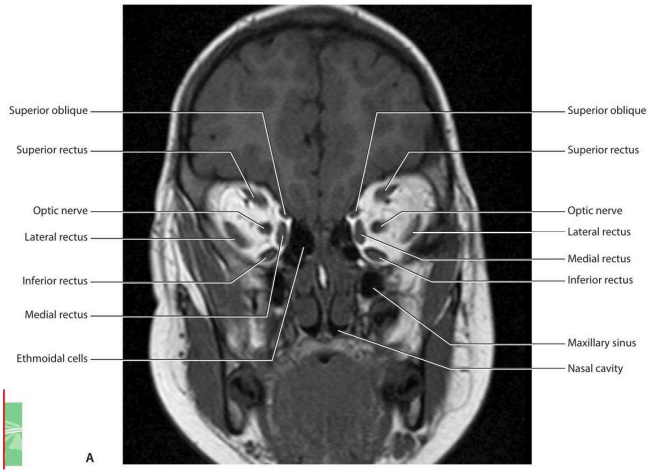
07 Radiological anatomy: On imaging

- **CT with contrast** for orbital complications: subperiosteal abscess is a lentiform collection against the lamina papyracea with displaced medial rectus.
- **MRI or CT venography** for cavernous sinus thrombosis: enlarged, non-enhancing sinus and dilated superior ophthalmic vein.

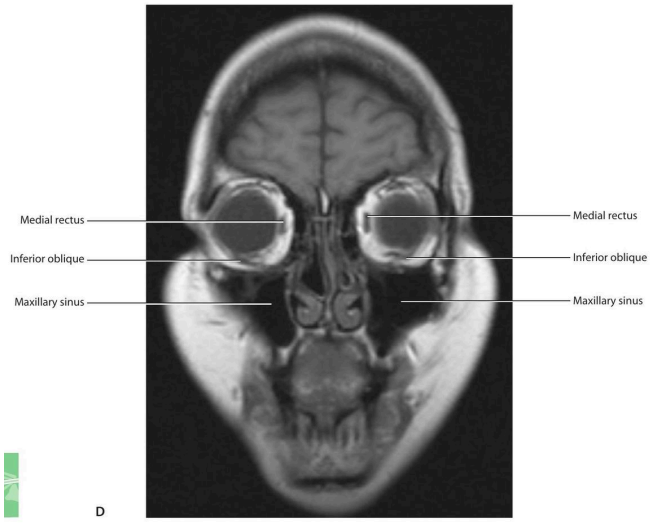


A through D – Coronal sections that pass through the orbit from posterior to anterior showing the extrinsic (extra-ocular) muscles and their relationships with each other and with other structures.
T1-weighted MR images in coronal plane

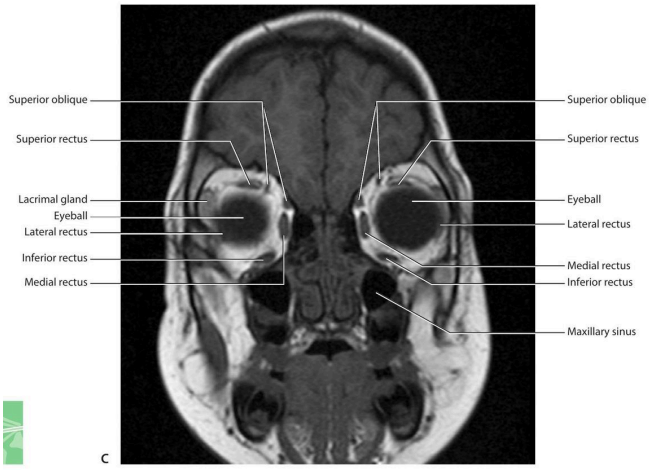
Coronal MRI through the orbit (posterior sections)
Gray's Atlas of Anatomy, 2nd ed., p. 518



Coronal MRI through the orbit: extra-ocular muscles and sinuses
Gray's Atlas of Anatomy, 2nd ed., p. 518



Coronal MRI through the anterior orbit
Gray's Atlas of Anatomy, 2nd ed., p. 519



Coronal MRI through the orbit: maxillary sinus relation
Gray's Atlas of Anatomy, 2nd ed., p. 519

08 Surgical anatomy: In theatre

- **Subperiosteal abscess:** drain endoscopically through the lamina papyracea after ethmoidectomy, or externally via a Lynch incision.
- **Lateral canthotomy and cantholysis** for orbital compartment syndrome: cut the lateral canthus, then the inferior crus of the lateral canthal tendon.
- **Endoscopic orbital decompression** removes the lamina papyracea and medial orbital floor (thyroid eye disease); keep a strut between the frontal recess and orbit.

T Tables

Fissures and canals of the orbit

Opening	Contents
Optic canal	Optic nerve, ophthalmic artery
Superior orbital fissure	III, IV, VI, V1 (lacrimal, frontal, nasociliary), superior ophthalmic vein
Inferior orbital fissure	V2 (infraorbital), zygomatic nerve, branches of the inferior ophthalmic vein
Anterior ethmoidal foramen	Anterior ethmoidal nerve and artery
Posterior ethmoidal foramen	Posterior ethmoidal nerve and artery
Nasolacrimal canal	Nasolacrimal duct
Infraorbital canal	Infraorbital nerve and vessels

Chandler classification

Stage	Name	Key signs	Usual management
I	Preseptal cellulitis	Lid oedema, normal vision and movement	Oral or IV antibiotics
II	Orbital cellulitis	Proptosis, chemosis, painful movement	IV antibiotics, nasal decongestion
III	Subperiosteal abscess	Globe displaced down and out	IV antibiotics ± drainage
IV	Orbital abscess	Severe proptosis, ophthalmoplegia, visual loss	Surgical drainage
V	Cavernous sinus thrombosis	Bilateral signs, III–VI palsies, sepsis	IV antibiotics, consider anticoagulation

Drain whenever vision is threatened, regardless of stage.

M Mnemonics

Chandler

Pre, Post, Sub, Abscess, Cavernous

Preseptal, orbital (post-septal) cellulitis, Subperiosteal, orbital Abscess, Cavernous sinus thrombosis

Cavernous sinus lateral wall

O TOM CAT

Oculomotor, Trochlear, Ophthalmic, Maxillary in the lateral wall; Carotid and Abducens inside the sinus

Ethmoidal foramina

24-12-6

Lacrimal crest → anterior ethmoidal 24 mm → posterior ethmoidal 12 mm → optic canal 6 mm

Q Rapid recall

1. **Thinnest wall of the orbit?** Medial wall (lamina papyracea).
2. **What passes through the inferior orbital fissure?** Maxillary nerve (to infraorbital), zygomatic nerve, branches of the inferior ophthalmic vein to the pterygoid plexus.
3. **Chandler stages of orbital complications?** I preseptal cellulitis, II orbital cellulitis, III subperiosteal abscess, IV orbital abscess, V cavernous sinus thrombosis.
4. **Rule of 24-12-6?** Anterior lacrimal crest to anterior ethmoidal foramen 24 mm, to posterior ethmoidal foramen 12 mm, to optic canal 6 mm (approximate).
5. **Order of nerves in the lateral wall of the cavernous sinus?** III, IV, V1, V2 from top to bottom.

6. **Which nerve lies inside the cavernous sinus beside the ICA?** Abducens (VI).
7. **Emergency treatment of retrobulbar haematoma?** Lateral canthotomy and inferior cantholysis.
8. **Where does the nasolacrimal duct open?** Inferior meatus, under the inferior turbinate, guarded by Hasner's valve.
9. **Most common sinus source of orbital complications in children?** Ethmoid sinuses.
10. **Sign of optic nerve compromise in orbital infection?** Reduced acuity, red desaturation and relative afferent pupillary defect.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.